

Henrique dos Santos Teixeira

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EXPERIENCE

Web Application Developer *Tuna de Ciências do Porto*

02/2025 – now | Porto, Portugal

- Co-developed and maintain a full-stack web application to manage events, members, instruments and organizational workflow for a student musical group, automating previously manual workflows.
- Informally act as technical lead, managing a team of 8 student contributors, handling onboarding, development workflow and self-hosting infrastructure with Cloudflare Tunnel and JWT authentication.
- Built with React (Vite), Go and MySQL, containerized with Docker Compose.

Researcher - AIMaCoV: ONNX Graph Mapping to RISC-V CGRA *INESC TEC*

02/2026 – 07/2026 | Porto, Portugal

- Proactively advocated to lead a new research initiative, mentoring an undergraduate intern on pipeline management and algorithmic strategy.
- Guided the integration of deterministic algorithms with GNNs to downscale the SAT solver search space, resulting in a 50x speedup.

Internship - LNS on a RISC-V SoC via High Level Synthesis *INESC TEC*

02/2026 – 06/2026 | Porto, Portugal

- Developed an API for LNS data formats with custom RISC-V instruction compilation, simulation-based types for arithmetic, comparisons and conversions between LNS and BFloat/Float.
- Validated 16-bit LNS arithmetic viability by executing a transformer-based LLM in simulation, achieving output text parity indistinguishable from float32 and bfloat16 baselines.

Researcher - AIQ-Ready: Accelerator for Compute-Heavy Nodes in ONNX Models *INESC TEC*

09/2025 – 01/2026 | Porto, Portugal

- Assessed viability of converting ONNX models to C via ONNX2C and profiled compute-heavy nodes in the resulting output.
- Selected bottleneck nodes for hardware acceleration on XHeep/RISC-V and evaluated combined SW + accelerated results.

Internship - AI Acceleration on a RISC-V SoC via High Level Synthesis *INESC TEC*

06/2025 – 07/2025 | Porto, Portugal

- Best CTM Internship – Computer Science and Engineering winner.
- Deployed a Logarithmic Number System Unit to a RISC-V CPU via HLS in C++, with addition/subtraction approximations using only 192 bytes of lookup tables combined.
- Integrated into the RISC-V pipeline and tested on FPGA at 125 MHz.

Academic Internship - Lambda Calculus Interpreter *FCUP*

02/2025 – 06/2025 | Porto, Portugal

- Implemented a C interpreter for Lambda Calculus compiled to SK Combinators, combining both theories to build translating and reducing functions over a custom program tree.

PAPERS

Logarithmic Number System Extension on an HLS-Based RISC-V SoC *INESC TEC (Under Review)*

2026

- Co-authored with Guilherme de Oliveira, José Paradela, Luís Miguel Sousa and Nuno Paulino.
- Designed a 16-bit LNSU for the RISC++ core achieving up to 45% fewer LUTs, 58% fewer DSPs and 29–36% higher clock frequencies vs. IEEE-754 FPU on a Zynq-7020 FPGA, with 95.99% MNIST accuracy and no retraining.

PROJECTS

Lightweight Cloud Job Platform on GCP

- CLI-driven platform on GCP submitting shell commands as jobs on auto-provisioned GCE worker VMs, with a FastAPI backend, Redis queue, autoscaler daemon (scale-to-zero), JWT auth, rate limiting and IAP tunneling.

LNS Library

- C++ library implementing 8-bit and 16-bit Logarithmic Number System arithmetic, with exact multiply/divide via exponent addition, spline-based LUT approximation for add/sub and both a hardware compiler header (custom RISC-V instructions) and a software simulation API for benchmarking against float32 and bfloat16.

RISC-V Assembler with Custom Instructions

- Assembler for RISC-V supporting 32IM instructions and custom LNSU instructions from INESC TEC internship.

Kotlin Compiler

- Compiler for a subset of Kotlin (statements, expressions, enums, functions) targeting MIPS assembly.

Memory Allocator Library

- Custom C library implementing an Arena (O(1) linear) and a Pool (best-fit, O(F)) allocator with coalescing, realloc and strdup support; used in the Kotlin Compiler and SK Interpreter.

EDUCATION

Licenciatura de Inteligência Artificial e Ciência de Dados *FCUP*

09/2023 – 06/2026 | Porto, Portugal

- 2× winner of “Bolsas PRR «Impulso Jovens STEAM» - Distinção Académica” scholarship.
- Relevant Courses: **Compilers** (Extra.), **AI**, **Machine Learning I & II**, **Data Structures**, **Algorithms**

TOOLS

Systems/Hardware: RISC-V, HLS, FPGAs, Docker

Infrastructure/Cloud: GCP, Redis, FastAPI, Cloudflare Tunnels, JWT

PROGRAMMING LANGUAGES

C | C++ | Python | Go | JavaScript